$(1) \times 1 \Rightarrow 2x - 3y = 6$

$(2) \times 2 \Rightarrow 2x + 2y = 2$

$-5y = 4 \Rightarrow y = \dfrac{-4}{5}$

Substituting $y = \dfrac{-4}{5}$ in (2), $x - \dfrac{4}{5} = 1$ we get, $x = \dfrac{9}{5}$

Therefore, $x = \dfrac{9}{5}$, $y = \dfrac{-4}{5}$.

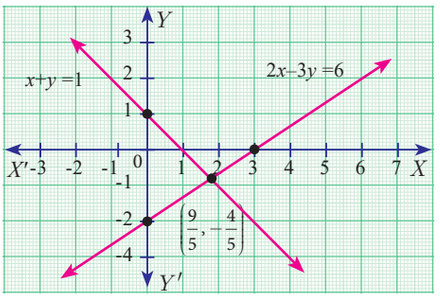


Fig. 3.1

## 3.2 Simultaneous Linear Equations in Three Variables

Right from the primitive needs of calculating amount spent for various items in a super market, finding ages of people under specific conditions, finding path of an object when it is thrown upwards at an angle, Algebra plays a vital role in our daily life.


Fig. 3.2

Any point in the space can be determined uniquely by knowing its latitude, longitude and altitude. Hence to locate the position of an object at a particular place situated on the Earth, three satellites are positioned to arrive three equations. Among these three equations, we get two linear equations and one quadratic (second degree) equation. Hence we can solve for the variables latitude, longitude and altitude to uniquely fix the position of any object at a given point of time. This is the basis of Global Positioning System (GPS). Hence the concept of linear equations in three variables is used in GPS systems.

### 3.2.1 System of Linear Equations in Three Variables

In earlier classes, we have learnt different methods of solving Simultaneous Linear Equations in two variables. Here we shall learn to solve the system of linear equations in three variables namely, $x$, $y$ and $z$. The general form of a linear equation in three variables $x$, $y$ and $z$ is $ax + by + cz + d = 0$ where $a$, $b$, $c$, $d$ are real numbers, and atleast one of $a$, $b$, $c$ is non-zero.

**Note**

- A linear equation in two variables of the form $ax + by + c = 0$, represents a straight line.

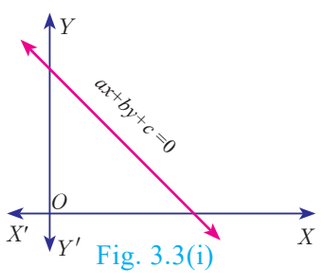


Fig. 3.3(i)

- A linear equation in three variables of the form $ax + by + cz + d = 0$, represents a plane.

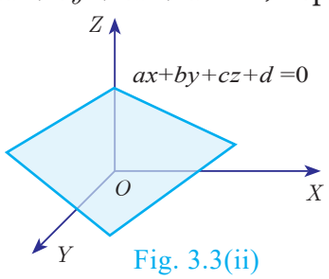


Fig. 3.3(ii)



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